

WASP-10b

R-2254

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_101120 · created 2026-08-16T00:30:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455520.72532 +/- 0.00083 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.059
Transit depth (Rp/Rs)^2	3.52 +/- 2.21 [%]
Orbital Inclination (inc)	87.95 +/- 1.91
Airmass coefficient 1 (a1)	0.62 +/- 0.27
Airmass coefficient 2 (a2)	0.28 +/- 0.42
Scatter in the residuals of the lightcurve fit is	38.41 %
Best Comparison Star	#2 - [243, 466]
Optimal Aperture	1.58
Optimal Annulus	4.13
Transit Duration (day)	0.057 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.188 $\pm$ 0.059 vs 0.1592 (deviation 0.45 σ)
Duration fit vs published	0.057 d vs 0.0946 d (deviation 3.16 σ)
Mid-time O-C	2.0 min
Depth SNR	2.9
Residual scatter	38.41 %

Light curve

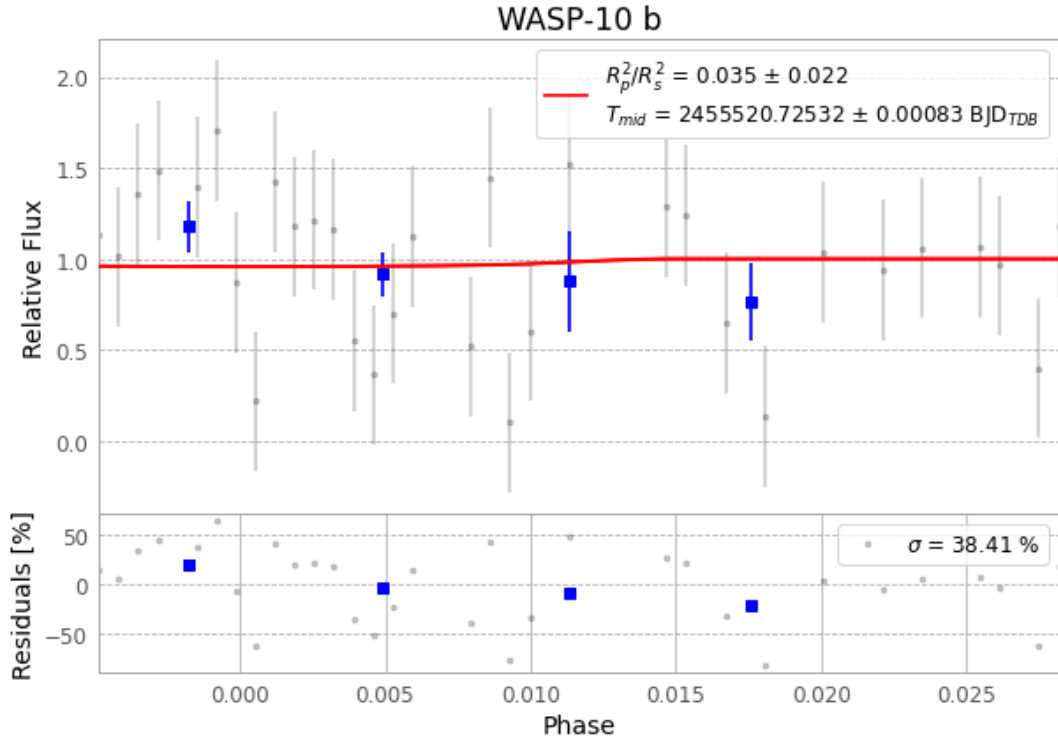


Figure 1 — FinalLightCurve_WASP-10 b_2010-11-19.png (SHA-256 dde0975918c6c44d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [241, 197], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 103470f7fc05230b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

50 FITS frames (33 MB), WASP-10101120045714.FITS → WASP-10101120072414.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-101120.log (79d929adb761...), FinalLightCurve_WASP-10 b_2010-11-19.png (dde0975918c6...), FinalLightCurve_WASP-10 b_2010-11-19.csv (6804d09770d6...)

Compute resources

Wall time 215.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 00:30Z.